

Dental Assistant Robot for Surgical Procedures

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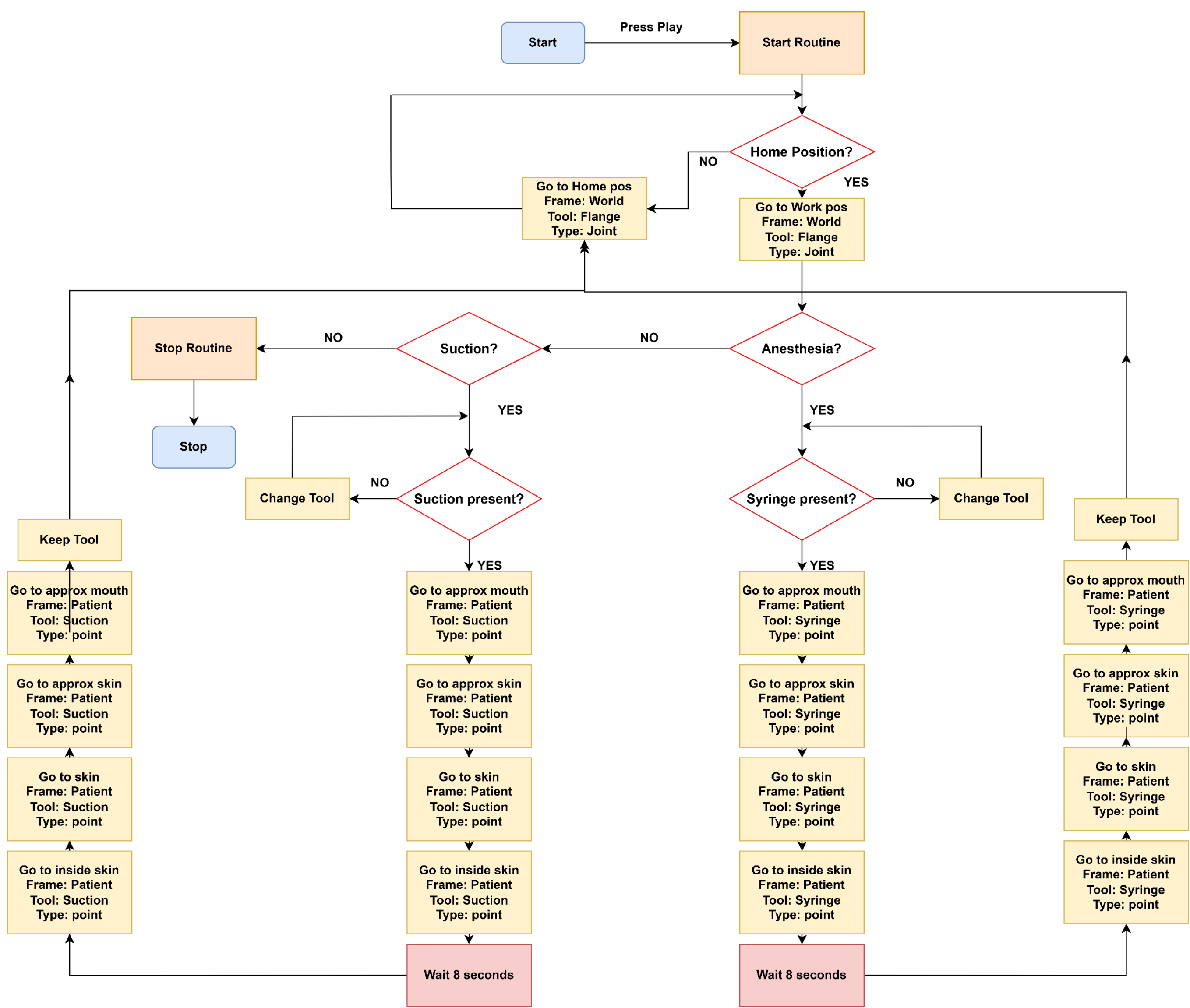
I. INTRODUCTION

Dental surgeries often involve repetitive and time-intensive tasks, such as administering anesthesia and performing suction, which are critical to ensuring patient comfort and procedural success. This project introduces a robotic system designed to assist in dental surgeries by automating key tasks during dental surgeries.

II. OBJECTIVE

- To develop a robotic system capable of performing anesthesia administration and dental suction with precision.
- To program the STAUBLI TX60 robot using the VAL3 programming language.
- To integrate user-friendly features and ensure adaptability for future advancements in robotic dentistry.

III. BLOCK DIAGRAM



IV. PROCEDURE

Working Procedure: Anesthesia



Home Position — Work Position — Assigning Tool — Work Position with Syringe — Performing Anesthesia

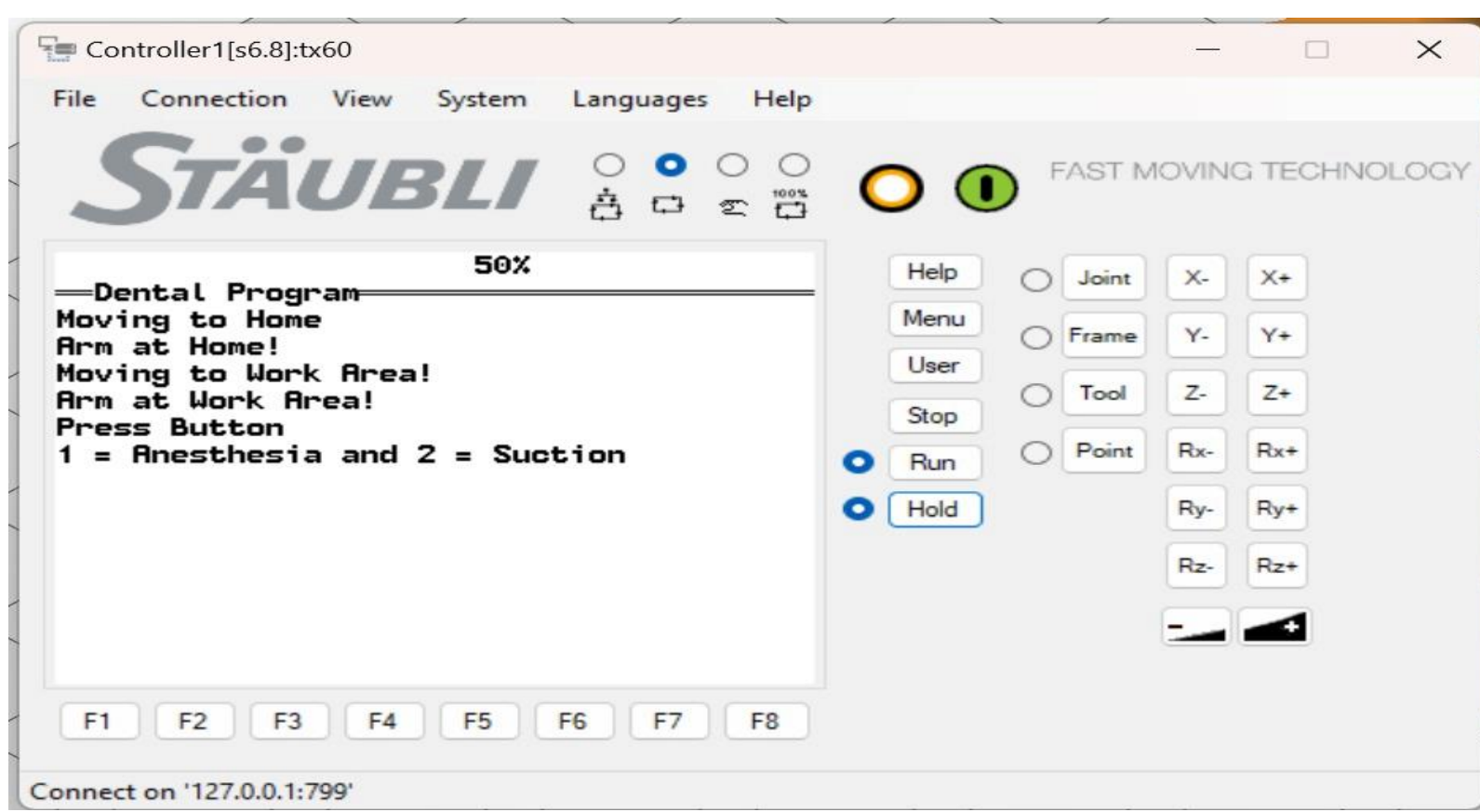
Working Procedure: Suction



Home Position — Work Position — Assigning Tool — Work Position with Suction — Performing Suction

V. INTERFACE

The interface shows the user steps of the procedure, and at the end the user decides whether to perform anesthesia or dental suction on the patient.



VI. LIMITATIONS

- The system was tested on a static skeleton model, whereas real-life applications require dynamic adjustments to account for patient movements like breathing or shifting positions.
- The use of fixed tools (syringe and suction) reduces adaptability compared to the varied tools commonly used in dental practices..
- The current setup lacks real-time imaging for mapping patient anatomy, which is critical for accurate application in real scenarios..

VII. FUTURE WORK

- **Orthodontics:** Assisting in wire placement and adjustments for braces, improving accuracy and reducing treatment time.
- **Oral Surgery:** Assisting in tasks like tissue retraction, suture placement, or surgical site cleaning to reduce surgical fatigue and improve workflow.
- **Implantology:** Providing support in implant placement during drilling and insertion.

